GAP-10I: Exact Augmented-Holonomy Criterion for a Prescribed Curved Branch

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Abstract

For prescribed smooth tetrad E_μ and paired connections (A_μ, B_μ), the equation $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$ is an inhomogeneous linear parallel-section problem. By adjoining one constant component it becomes a homogeneous problem for an augmented connection. This yields an exact local/global criterion: a solution through initial value Θ_0 exists and is path independent precisely when the augmented holonomy fixes $(\Theta_0, 1)$; flat augmented curvature is a simple sufficient condition for all initial values. The curvature equation reproduces the two-sided intertwiner condition in the torsion-free case. This closes the prescribed- coefficient integrability problem, while the genuinely UBT-specific self-consistent selection of E, A, B remains open.

The augmentation of an inhomogeneous linear parallel-transport problem by one constant component, together with the resulting holonomy criterion, is standard affine-connection and parallel-transport machinery; see, for example, Kobayashi–Nomizu [1]. The UBT-specific content here is its application to the paired biquaternionic equation and the exact block curvature identity below.

1 Inhomogeneous paired system

Let $\mathcal{V} = \mathbb{B} \simeq \mathrm{Mat}(2,\mathbb{C})$ and define the connection operator

$$\mathcal{C}_\mu X := A_\mu X - X B_\mu. \tag{1}$$

The tetrad-generation equation is

$$\partial_\mu\Theta + \mathcal{C}_\mu\Theta = sE_\mu, \qquad s := \sqrt{\mathcal{N}_0}. \tag{2}$$

Introduce the augmented vector $Y = (\Theta, 1)^T \in \mathcal{V} \oplus \mathbb{C}$ and connection

$$\widehat{\mathcal{C}}_\mu = \begin{pmatrix} \mathcal{C}_\mu & -sE_\mu \\ 0 & 0 \end{pmatrix}, \tag{3}$$

where the upper-right entry maps a scalar z to $-szE_\mu$. Then Eq. (2) is exactly

$$\widehat{D}_\mu Y := (\partial_\mu + \widehat{\mathcal{C}}_\mu)Y = 0. \tag{4}$$

2 Curvature and compatibility

Define

$$\mathcal{F}_{\mu\nu}X := F^A_{\mu\nu}X - XF^B_{\mu\nu}, \tag{5}$$

$$\mathcal{D}_\mu E_\nu := \partial_\mu E_\nu + A_\mu E_\nu - E_\nu B_\mu. \tag{6}$$

Theorem 2.1 (Augmented-curvature identity)**.** *The curvature of Eq.* (3) *is*

$$\boxed{\widehat{\mathcal{F}}_{\mu\nu} = \begin{pmatrix} \mathcal{F}_{\mu\nu} & -s(\mathcal{D}_\mu E_\nu - \mathcal{D}_\nu E_\mu) \\ 0 & 0 \end{pmatrix}.} \tag{7}$$

Consequently every solution obeys

$$\boxed{F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu} = s(\mathcal{D}_\mu E_\nu - \mathcal{D}_\nu E_\mu).} \tag{8}$$

For torsion-free antisymmetric tetrad compatibility, the right-hand side vanishes and Eq. (8) *reduces to the curvature intertwiner condition.*

Proof. Multiply the block matrices in $\partial_\mu\widehat{\mathcal{C}}_\nu - \partial_\nu\widehat{\mathcal{C}}_\mu + [\widehat{\mathcal{C}}_\mu, \widehat{\mathcal{C}}_\nu]$. The upper-left block is $\mathcal{F}_{\mu\nu}$; the upper-right block is the stated covariant curl. Applying $\widehat{\mathcal{F}}_{\mu\nu}$ to the parallel vector $(\Theta, 1)^T$ gives Eq. (8). □

3 Exact holonomy criterion

Theorem 3.1 (Prescribed-coefficient existence and path independence)**.** *Let U be connected, choose $x_0 \in U$ and $Y_0 = (\Theta_0, 1)^T$. Parallel transport of Eq.* (4) *along every piecewise smooth path from x_0 exists uniquely. It defines a single-valued solution on U if and only if the holonomy group of $\widehat{\mathcal{C}}$ at x_0 fixes Y_0:*

$$\mathrm{Hol}_{x_0}(\widehat{\mathcal{C}})Y_0 = Y_0. \tag{9}$$

When this holds, the solution is unique for the initial value Θ_0.

Proof. Along a path γ, Eq. (4) is a linear ordinary differential equation and has unique path-ordered-exponential transport. Two paths with the same endpoints give the same value exactly when transport around their closed concatenation fixes Y_0. This is Eq. (9). Uniqueness along paths then gives uniqueness of the resulting section. □

Corollary 3.2 (Flat augmented connection)**.** *If U is simply connected and $\widehat{\mathcal{F}}_{\mu\nu} = 0$, then for every initial Θ_0 there is a unique global solution on U. More generally, nonzero curvature is allowed provided its holonomy stabilizes the selected augmented initial vector.*

Remark 3.3. The criterion is stronger and more precise than saying merely that mixed partial derivatives must agree. It also separates local/pathwise solvability, which is automatic, from multidimensional path independence and global single-valuedness, which are controlled by augmented holonomy.

4 Status

Defensible status

GAP-10I-PRESCRIBED: CLOSED [L1]. For specified smooth (E_μ, A_μ, B_μ) and an initial value, the exact existence, uniqueness and global path-independence criterion is the augmented holonomy condition Eq. (9); Eq. (8) is its curvature-level necessary condition.

GAP-10I-CURVED: LOCAL KINEMATICS CLOSED; DYNAMICS/GLOBAL PART NARROWED. The prescribed-coefficient PDE has the exact criterion above, and the companion construction in `gap_10i_torsionful_local_representer.tex` provides an explicit local single-Θ right inverse for every smooth tetrad using composite metric-compatible contortion. What remains is self-consistent and dynamical: derive the selected torsion/connection from the same UBT action, verify physical admissibility and the holonomy condition, and establish regularity and global continuation.

References

[1] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vol. I, Wiley, New York (1963).